SMART STUDENT

Student Management & Task Tracking System

Plan • Track • Achieve

Student	Md. Mashrafi Rahman
ID	IT-24034
Course	SEPM-ICT-3107
Project Type	Software Engineering Project / Initial Prototype

1. Project Idea

Smart Student Management & Task Tracking System is a centralized academic task-management platform. It allows students to manage their courses, academic tasks, assignments, deadlines and progress from one interface. Teachers can manage courses and assignments, while an admin can manage users and system data.

2. Problem Statement

Students often manage courses, assignments, deadlines and academic tasks using different applications, notes or manual lists. This can make it difficult to identify pending work, upcoming deadlines and overall academic progress. The proposed system provides a centralized solution for organizing these activities.

3. Objectives

- Provide a centralized student task-management system.
- Allow students to manage their courses and academic tasks.
- Track assignment deadlines and task priorities.
- Allow tasks to be marked as completed or pending.
- Provide an overview of upcoming work and academic progress.
- Allow teachers to manage courses and assignments.
- Reduce missed academic deadlines.

4. Main Users

User	Main Responsibilities
Student	Login, view dashboard, manage courses, create/update/delete tasks, view assignments, track deadlines ar
Teacher	Manage courses, create assignments, set deadlines and view student progress.
Admin	Manage users, courses and system data.

5. Core Features

Module	Features
Authentication	Login, logout and role-based access.
Course Management	Add, update, delete and view courses.
Task Management	Create, update, delete, complete tasks and set priorities.
Assignment Management	Create assignments, set deadlines and track status.
Dashboard	Courses, pending tasks, completed tasks, upcoming deadlines and progress.

6. Software Process Model — Agile Scrum

The project will follow the Agile Scrum software process model. Development will be divided into small iterations so that requirements and features can be reviewed and improved throughout the project.

```
Product Backlog → Sprint Planning → Development → Testing → Sprint Review → Feedback → Next Sprint
```

7. Project Schedule

Week	Major Activity
1	Requirement Analysis
2	Proposal & Requirement Specification
3	System Design & UML
4	Database Design
5	UI Design
6	Core Development
7	Feature Development
8	Integration & Testing
9	Bug Fixing & Improvements
10	Documentation & Final Submission

8. Milestones

M1	Requirements Completed
M2	Proposal Completed
M3	UML Design Completed
M4	Database Design Completed
M5	UI Prototype Completed
M6	Core System Completed
M7	Testing Completed

M8	Final Documentation Completed
M9	Final Submission

9. Gantt Chart — Initial Plan

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Requirements	■	■								
Proposal		■	■							
UML Design			■	■						
Database				■	■					
UI Design					■	■				
Development						■	■			
Testing								■	■	
Documentation									■	■

10. System Flow

The overall user flow is designed around a simple academic-management workflow:

```
LOGIN → DASHBOARD → COURSES / TASKS / ASSIGNMENTS → PROGRESS
```

11. Data Flow Diagram — Concept

The DFD will show how information moves between the users, system processes and stored data. A student can submit task information to Task Management, which stores or retrieves task data and returns the result to the student.

```
Student → Task Management → Database → Task Information → Student
```

```
Teacher → Assignment Management → Database → Assignment Information → Student
```

12. Use Case Diagram — Main Use Cases

Actor	Main Use Cases
Student	Login; View Dashboard; Manage Courses; Create/Update/Delete Task; Complete Task; Set Priority; View Ass
Teacher	Login; Manage Course; Create Assignment; Update Assignment; Set Deadline; View Student Progress.
Admin	Login; Manage Users; Manage Courses; Manage System Data.

13. Activity Diagram — Add Task

```
START → Login → Dashboard → Add Task → Enter Information → Validate → Valid? → (No: Show
Error → Enter Again) / (Yes: Save Task → Show Success → END)
```

14. Entity Relationship Diagram — Main Entities

The database design will contain the following main entities and relationships:

```
USER → STUDENT / TEACHER
STUDENT → ENROLLMENT → COURSE
STUDENT → TASK
TEACHER → ASSIGNMENT
ASSIGNMENT → SUBMISSION ← STUDENT
```

15. Class Diagram — Proposed Classes

Class	Purpose
User	Common user information and authentication-related data.
Student	Student identity and academic functions.
Teacher	Teacher identity and assignment/course management.
Course	Course information.
Task	Student-created academic tasks, priority and status.
Assignment	Teacher-created academic assignments and deadlines.
Submission	Student submission information.
Notification	Deadline or system notification information.

16. Object Diagram — Example

```
student01 : Student
studentId = IT-24034
name = Md. Mashrafi Rahman

course01 : Course
courseId = SEPM-ICT-3107
courseName = Software Engineering

task01 : Task
title = UML Diagram
priority = HIGH
status = PENDING
```

17. User Interface Concept

The initial interface will use a clean Dark Academic Tech theme. The main colors are deep navy, royal blue, teal, white/light gray, amber for deadlines and red for overdue items.

Screen	Main Elements
Login	Username, password, Login button.
Dashboard	Welcome message, course count, pending tasks, completed tasks, upcoming tasks.
Courses	Course list, add/update/delete course.
Tasks	Task list, priority, status, deadline, add task.
Assignments	Assignment title, course, deadline and status.

18. Java Prototype — Minimum Classes

The initial prototype can be implemented using Java, with at least five classes from the class diagram. A manageable first version can use Java Swing for the interface.

```
User
Student
Teacher
Course
Task
Assignment
```

Suggested UI classes: LoginFrame, DashboardFrame, CoursePanel, TaskPanel and AssignmentPanel.

19. Testing Plan

Test Case	Expected Result
Correct login	Dashboard opens successfully.
Incorrect login	Error message is displayed.
Create task	New task appears in the task list.
Update task	Updated information is displayed.
Complete task	Task status changes to Completed.
Create assignment	Assignment is visible to students.
Deadline display	Upcoming deadline is shown on dashboard.

20. Deliverables

- Project Proposal
- Requirements Specification
- Software Process Model
- Project Schedule and Gantt Chart
- Data Flow Diagram
- Use Case Diagram

- Activity Diagram
- Entity Relationship Diagram
- Class Diagram
- Object Diagram
- Java UI Prototype
- Source Code
- Testing Documentation
- Final Project Documentation
- GitHub Repository

21. Proposed GitHub Structure

```
Smart-Student-Management-System/
■■■ README.md
■■■ Proposal/
■ ■■■ Project-Proposal.pdf
■ ■■■ Project-Proposal.tex
■■■ UML/
■ ■■■ DFD.png
■ ■■■ Use-Case.png
■ ■■■ Activity.png
■ ■■■ ERD.png
■ ■■■ Class-Diagram.png
■ ■■■ Object-Diagram.png
■■■ Gantt/
■ ■■■ Gantt-Chart.png
■■■ src/
■■■ ui/
■■■ screenshots/
■■■ docs/
```

22. Final Project Scope

The first version intentionally keeps the implementation manageable. The core prototype focuses on login, dashboard, courses, tasks, assignments and progress. A full production backend, advanced database security, cloud deployment and mobile application can be considered future extensions rather than requirements of the initial prototype.

23. Final Explanation

The complete development flow is: Idea → Requirements → Agile Planning → Gantt Schedule → UML/System Design → Java Prototype → Testing → Documentation → GitHub Submission. The objective is not to build a large commercial application immediately, but to demonstrate a complete software-development process with a coherent proposal, design and working initial interface.